

Capstone failure testing

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone service implementation

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Red team exercises

Without opening the old lesson, explain one key idea from Red team exercises. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Accessibility and user feedback

Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Capstone failure testing. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Capstone failure testing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

For Grounded Learning Service, use this week's independent task as a rehearsal for the final evidence pack: Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Capstone failure testing

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

F4 records club time 15:00. The contract permits a checked fixed-template answer when drafting is unavailable.

Capstone failure testing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Verify source

Retrieval supplies F4 and the time 15:00.



2. Inject failure

Disable drafting without changing source evidence.



3. Use bounded fallback

Return the checked template citing F4 and record degraded operation.

CHANGE ONE THING

Make retrieval unavailable as well.

Explain the changed result using the matching lesson.

Explain why retrieval success permits the checked 15:00 fallback when drafting fails. What changes if retrieval fails too, and why can a template not fill that evidence gap?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Return the checked template citing F4 and record degraded operation.

1. Verify source
2. Use bounded fallback
3. Inject failure

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Use bounded fallback Return the checked template citing F4 and record degraded operation. Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain evidence checks and honest status.

Practice 2: explain the deciding evidence

Changed-case practice: Make retrieval unavailable as well. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Return unsupported or unavailable; do not use the factual template.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Return unsupported or unavailable; do not use the factual template. A template can phrase evidence but cannot replace missing evidence.

Repair a missing building block

Review Capstone service implementation (ai-6/33). Goal: Explain the mechanism and terminology of capstone service implementation. Short practice: Trace the G2 fact request and edit request through the prototype. What can be returned, what cannot be done, and how does removing G2 change those two outcomes?. Return to this week and explain what you changed.

Review Red team exercises (ai-6/31). Goal: Explain the mechanism and terminology of red team exercises. Short practice: Distinguish the exhibit fact from the embedded edit instruction. Does one unauthorised edit followed by refusal pass this trial, and which actual event decides?. Return to this week and explain what you changed.

Review Accessibility and user feedback (ai-6/28). Goal: Explain the mechanism and terminology of accessibility and user feedback. Short practice: Explain what each text, keyboard and assistive-technology change adds to the feedback. Why does changing the red circle to orange alone not solve the communication barrier?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
